

DIY Robot Challenge 2026 - Installation Setup

Generated on 2026-05-16 for workspace: **DIY-Challenge-Repo**

1. Base Platform

- Ubuntu 22.04 LTS
- ROS 2 Humble
- colcon and rosdep

2. APT Packages Installed

- ros-humble-robot-localization
- ros-humble-navigation2
- ros-humble-nav2-bringup
- ros-humble-pcl-ros
- ros-humble-realsense2-camera
- ros-humble-nmea-navsat-driver
- ros-humble-ublox-gps
- ros-humble-octomap-server
- ros-humble-apriltag-ros
- ros-humble-gtsam
- libdw-dev, libelf-dev

3. Source Workspaces and Packages

3.1 Repo-local source workspace

`third_party_ws` inside the main repo contains source packages:

- livox_ros_driver2
- fast_lio
- lio_sam
- code_utils
- imu_utils
- ndt_omp_ros2
- lidar_imu_calib

3.2 Additional source-built tools

- micro_ros_agent in `~/micro_ros_ws`
- hesai_ros_driver in `~/ros2_ws/src/Betsybot-Software/src/HesaiLidar_ROS_2.0`
- Livox-SDK2 library installed at `/usr/local/lib/liblivox_lidar_sdk_shared.so`

4. Environment Setup (Runtime)

Source these in every new terminal before launch:

```
source /opt/ros/humble/setup.bash
source ~/micro_ros_ws/install/setup.bash
source ~/ros2_ws/src/Betsybot-Software/src/HesaiLidar_ROS_2.0/install/setup.bash
source ~/ros2_ws/src/DIY-Challenge-Repo/third_party_ws/install/setup.bash
```

5. Verification Commands

```
ros2 pkg prefix hesai_ros_driver
ros2 pkg prefix micro_ros_agent
ros2 pkg prefix livox_ros_driver2
ros2 pkg prefix fast_lio
ros2 pkg prefix lio_sam
ros2 pkg prefix robot_localization
ros2 pkg prefix nav2_bringup
ros2 pkg prefix pcl_ros
ros2 pkg prefix realsense2_camera
ros2 pkg prefix nmea_navsat_driver
ros2 pkg prefix ublox_gps
ros2 pkg prefix octomap_server
ros2 pkg prefix apriltag_ros
ros2 pkg prefix imu_utils
ros2 pkg prefix ndt_omp_ros2
ros2 pkg prefix lidar_imu_calib
```

6. Build Stability Notes

- Used conservative build settings to prevent terminal crashes: `--executor sequential --parallel-workers 1` and `MAKEFLAGS=-j2`.
- Some upstream third-party packages required compatibility fixes for Humble builds.

Status: Required packages from the SAD are installed and discoverable in the configured overlays.

Tip: Keep `third_party_ws/src` under version control if you want reproducible team setups.